

Multiple Choice (10 marks)

1. What is the least common multiple of 4 and 10?
 - (a) 100
 - (b) 20
 - (c) 14
 - (d) 40
 - (e) 80
2. A car gets 24 miles per gallon of gasoline (mi/gal). How many gallons of gasoline would the car need to travel 144 miles?
 - (a) 5 gallons
 - (b) 6.2 gallons
 - (c) 7.5 gallons
 - (d) 7 gallons
 - (e) 5.8 gallons
3. What is the least possible positive integer-value of n such that $\sqrt{18 \cdot n \cdot 34}$ is an integer?
 - (a) 4
 - (b) 3
 - (c) 1
 - (d) 18
 - (e) 51
4. Solve the following linear programming problems: maximize $3x + y$ subject to (1) $-x + y \leq 1$, (2) $2x + y \leq 4$, (3) $x \geq 0$ and $y \geq 0$. What's $[x, y]$ for the optimal solution?
 - 1, 2
 - 0, 1
 - 1, 1
 - 4, 0
 - 3, 0
5. If $3(x - 3) + 10 = 19$, then $x =$
 - (a) 4

- (b) 1
- (c) 9
- (d) 3
- (e) 2

True / False (10 marks)

Answer True or False

6. True or False: The number of ways to seat 8 people at 2 identical round tables is 40320, assuming the tables are distinct.
7. True or False: The probability $P(A \cup B)$ for independent events A and B is equal to $P(A) + P(B) = 1/4 + 2/3 = 11/12$.
8. True or False: If Pat bounces a basketball 25 times in 30 seconds, he will bounce it approximately 200 times in 150 seconds.
9. True or False: The solution to the equation $x - 10.5 = -11.6$ is $x = 10.5$.
10. True or False: The value of the expression $14 + (-71)$ is -43.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. In the triangle shown, what is the positive difference between the greatest and least possible integral values of x ?
`[asy] defaultpen(linewidth(0.7)); pair a,b,c; b = (1,2); c = (3,0); draw(a--b--c--a); label("x",a--b,dir(135)); label("5",b--c,dir(45)); label("6",c--a,S); [/asy]`
12. Nine tiles are numbered $1, 2, 3, \dots, 9$, respectively. Each of three players randomly selects and keeps three of the tiles, and sums those three values. The probability that all three players obtain an odd sum is m/n , where m and n are relatively prime positive integers. Find $m + n$.

End of Paper